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THE VOWELING OF "i TYPE" SEGOLATES IN TIBERIAN HEBREW

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THE history of the development of *i* in segolate nouns, i.e., nouns of original base form *CiCC, remains obscure. This paper presents a new attempt to understand the development of the vowel, resulting from continuing study of cases of apparent inconsistency in vowel or stress patterns in the Tiberian tradition of biblical Hebrew.¹

1. Scholars disagree on the form which the normal linguistic processes should have produced. P. Joüon states that the *i* vowel should regularly have given rise to *šere*, as in *šēfer*.² H. Bauer and P. Leander argue that the *i* should have changed to *a* in all cases under Philippi's law but that this change has often been prevented.³

2. Even the identification of nouns of original form *CiCC presents something of a problem. The following criteria are used here. If, in a segolate noun, the vowel following the first consonant is:

- (a) *šere* where the syllable is open and stressed in context, as *šēfer*, Isa. 30:8,
- (b) *šere* or *segol* where the syllable is open and stressed in pausal situations,⁴ as *šēfer* (Jer. 30:2), *heṭah* (Gen. 34:25),
- (c) *hireq* or *segol* where the syllable is closed and unstressed, as before suffixes in *sifrî*, *negdô*, then the original vowel is assumed to have been *i*.⁵ However, only a few forms show all these characteristics, and it is possible that they do not always reflect original *i*. Problems are, under (a) (the syllable is open and stressed in context): short *i* has given rise to *segol* in many segolates, as *šedeq*, and has almost always done so in segolate endings in longer nouns, as feminine singular *qal* participles; (b) (the syllable

¹ I was able to devote full time to this research as a fellow of the Institute for Advanced Studies at the Hebrew University in Jerusalem in 1983–84. I am most grateful to the directorate of the Institute and its staff for providing me with this opportunity and the excellent facilities in which to carry on my work. This paper, a partial result of that work, is preliminary to others in which the significance of the relation of a word to its context ("close-joining" or separation, see §8.1) is further demonstrated.

² P. Joüon, *Grammaire de l'Hébreu biblique*, 2d ed. (Rome, 1947). Variations from the "normal" *šere* in context or pause are due to the influence of the *CaCC pattern (§96Ae, f). See also the short statement by J. Blau in "On Pausal Lengthening, Pausal Stress Shift, Philippi's Law and Rule Ordering in

Biblical Hebrew," *Hebrew Annual Review (HAR)* 5 (1981): 8–9.

³ H. Bauer and P. Leander, *Historische Grammatik der hebräischen Sprache des Alten Testaments* (Halle, 1922) (hereafter cited as Bauer-Leander). The expected change is prevented through "Systemzwang" (§61a) or "Analogie" (§72e).

⁴ The term "pausal situation" in this paper indicates that the word in the text is marked with *silluq* or *atnah*, or, in the three "poetical" books, *oleh we-yored*. This is the only objective way in which the situations in which pausal forms are to be expected can be identified, but it should be noted that only about 80 percent of pausal forms occur in these situations. See my article "Pausal Forms and the Structure of Biblical Poetry," *VT* 31 (1981): 187, ¶4.

⁵ The information used here was obtained from A. Even-Shoshan, *A New Concordance of the Bible* (Jerusalem, 1982), save that the vowelings of the Leningrad Manuscript is followed in the few cases where it differs. Even-Shoshan's listing of homographs is also followed.

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(c) *segol* occurs in ḥbl , ḥsl , ḥr "side," ḥgl , ḥdr "flock," ḥzr "help," ḥrk , and also in ḥgly Job 38:28. *Hireq* occurs in ḥst , ḥmq , ḥsb .

3.2 After initial *h* or *s*, the reflex is usually (a) *ṣere* (*segol*), (b) *ṣere* (*qameṣ*), (c) after *s*, *hireq*, after *h*, *segol* (*hireq*, *pataḥ*).

(a) *Ṣere* occurs after *h* in twenty-three nouns, after *s* in four.¹² *Segol* occurs in *ḥbl* "cord," *ḥdr*, *ḥld* "world." Both vowels occur in *sml* (*ṣere* in construct, *segol* elsewhere) and in *ḥls* in some sources but not in L.

(b) *Ṣere* occurs in *ḥlq*, *ḥmt*, *ḥps*, *ḥpr* PN city, *ḥqr*, *ḥrm* "devoted thing" and "net," and *spr*. *Qameṣ* occurs in *ḥdq*, *ḥdr*, *ḥld* "world," *ḥls*, *str*, *sml*.

(c) *Segol* occurs in *ḥbl* "pain," *ḥt*, *ḥlb*, *ḥlq* "portion," *ḥps*, *ḥrm* "devoted thing" and "net," and also *ḥld* "world." *Hireq* occurs in *ḥqr*, *ḥsq*, and *sbl*, *spr*, *str*. Both *segol* and *pataḥ* occur in *ḥbl* "cord," *ḥdr*, and *ḥls*.

3.3 After initial *b*, *d*, *t*, *q*, *l*, the reflex is (a) *segol*,¹³ (b) *qameṣ* or *segol*, (c) *hireq* (*segol*, *pataḥ*).

(b) *Qameṣ* occurs after *b*, *d*, *t*, except where the final consonant is ḥ or, in one case, *ḥ*.¹⁴ *Segol* occurs in these cases, and also after *l* in *ldt* and *lqh*, and in *lkt* in six cases but not in three others, which show *qameṣ*.¹⁵ After *q*, *qameṣ* occurs in *qbr*, *qsp*, *qsr*, *segol* in *qdm*, *qds*, *qrb*.

(c) *Segol* occurs only in *lkt*. *Pataḥ* occurs in *tn* (Deut. 28:5, 17) and *qdm* (Prov. 8:23). All other forms show *hireq* except *ḥgd* "treachery," *ḥth* "security," *dš*, and *qds*, which do not occur in this situation. Two forms in this group show the reflex in a stressed closed nonfinal syllable (before *he* directional), *segol* in *qds*, *ṣere* in *qdm*.

3.4 After initial *g*, *p*, *ṣ*, and *r*, the reflex is usually (a) *segol* (*ṣere*), (b) *qameṣ* (*segol*), (c) *hireq*.

(a) *Segol* occurs after *g* in four nouns, after *p* in eleven, after *ṣ* in three, and after *r* in eight.¹⁶ *Ṣere* occurs in *gz* (Isa. 1:11), *gzl* (Ezek. 18:18, Qoh. 5:7), *pšr* (Qoh. 1:8), *pth* "opening" (Ps. 119:130), and *rbs* (Isa. 65:10), all of which are construct. Both vowels occur in *sl* (always construct): *segol* nine times, *ṣere* in 2 Sam. 16:13.

(b) *Qameṣ* occurs after *g* in *gšm* "rain," after *p* in *pgr*, *pšl*, *pš*, *prh*, *prš*, *pš*, *pth* "doorway," also *pdr*, which does not occur in situation (a). *Segol* occurs in *gh*, *pl*, and also after *ṣ* in *sdq*, *smh*. After *r*, *qameṣ* occurs in *rkb*, and *ršt* both "net" and the infinitive, and in *hrš* (Qoh. 3:16). *Segol* occurs in *rdt* and *rš* (thirteen cases without the article).¹⁷ In Gen. 19:6, *pth* "doorway" shows *segol* where the syllable is closed and stressed in pause.

¹² *ḥbl* "pain" and PN, *ḥdq*, *ḥlb* "fat" and PN, *ḥld* PN, *ḥlm* PN, *ḥlp* "exchange" and PN, *ḥlq* "smoothness" "portion," and PN, *ḥmt*, *ḥps*, *ḥpr* PN city, PN person, *ḥps*, *ḥqr*, *ḥrm* "devoted thing" and "net," *ḥsb*, *ḥsq*, and also the monosyllable *ḥt*, *sbl*, *spl*, *spr*, *str*.

¹³ *ḥgd* "garment," *ḥdq*, *ḥth* "security," *ḥtn*, *ḥs*, *brk*, *dgl*, *dš*, *dšn*, *ṭbh* "slaughter," *tn*, *qbr*, *qdm*, *qds*, *qrb*, *qsp*, *qsr*, *ldt*, *lkt*, *lqh*.

¹⁴ *Qameṣ* in *ḥgd* "treachery" and "garment" *ḥdq*, *ḥtn*, *ḥs*, *dšn*, *ṭbh* "slaughter"; also, in forms not found in situation (a), *hl*, and (if Even-Shoshan's homographs derive from the same original, as seems likely; cf. F. Brown, S. R. Driver, and C. A. Briggs, *A Hebrew and English Lexicon of the Old Testament* [1907; Oxford, 1968] [BDB]), *ḥvr*. *Segol* occurs in

ḥth "security," *dš*, *tn*.

¹⁵ *Segol* in Exod. 8:24; Judg. 19:5, 7; 1 Sam. 15:27; Ezek. 1:20; Ps. 78:10; *qameṣ* in Prov. 15:21, 30:29, Qoh. 1:7.

¹⁶ *gh*, *grš*, *gšm* "rain," *gšt*, *pgr*, *pl*, *plk*, "district," *pšl*, *pš*, *prh*, *prš* "breach," *prš* "excrement," *pš*, *pth* "doorway," *sdq*, *smh*, *rb* "quarter," *rdt*, *rkb*, *rsn* "halter," *rš*, *ršp*, *ršt* "net" and the infinitive. Both Even-Shoshan and BDB treat *petah* "doorway" and *petah* "opening" as independent, but they probably derive from a single original. If so, *pth* should be classed as a form which shows both vowels.

¹⁷ 1 Sam. 24:13, Isa. 58:4, Ezek. 7:11, Mic. 6:10, 11, Ps. 45:8, 84:11, Prov. 4:17, 8:7, 10:2, 12:3, 16:12, Job 34:8.

a mid-front vowel, "tonal lengthening") occurs after gutturals and *s*, the maximum change (shift to a central low vowel) after voiced stops and most other sounds, in pause if not in context.

4.2 In nouns of original form **CiC*;²⁴ *i* gives rise to *pataḥ* after *b*, *p*, *s*. The vowel is usually the same both in pause and elsewhere, but *qameṣ* occurs in a few cases.²⁵ *Ṣere* occurs after *ḥ*, [◌]*ḥ*, *k*, *n*, *s*, *t*.²⁵ *Ṣere* also usually occurs after [◌] ([◌]*ḥ*, [◌]*m*, [◌]*t* "with" and "tool" but *pataḥ* in [◌]*t*) and after *q* (*qṣ* and usually *qn*, but *pataḥ* appears in *qn* in the construct in Deut. 22:6). Either vowel may appear after *g*, *m*, *s*: *pataḥ* in *gt* ("wine-press" and the city), *mṣ*, *ṣd*; *ṣere* in *gz*, *mṣ*, *ṣl*. Here the development is clearly somewhat different from nouns of form **CiCC*.²⁶ However the minimal change ("tonal lengthening") occurs again after gutturals (and also after most other sounds). The greatest change (shift to a low central or back vowel) occurs after voiced stops (although not always), after *m*, *s*, *q*, and, surprisingly, even after [◌] and *s*.

4.3 Apocopated imperfect *qal* forms from roots III *h* originally formed doubly closed monosyllables as did nouns of form **CiCC*. The initial consonant in these forms can only be [◌], *y*, *n*, or *t*, but in a few cases these consonants do appear to affect the development of the following vowel in different ways. These forms also show instances where the development of the vowel is affected by the second or third consonant (i.e., the first or second consonant of the verbal root). The *i* of the prefix in these forms²⁷ gives rise to

qameṣ in three cases in pause,²⁸ and also apparently in *wṣr* Hos. 12:5.

pataḥ in roots *lh* and *lḥ*, and in the monosyllabic form *wṣr*.

Ṣere in roots *ll*, [◌]*h*, [◌]*ḥ* (except for [◌]*ḥ*); in most monosyllabic forms (from *bkh*, *nth*, *rdh*, *ṣth*, and also *wṣz* Isa. 63:3); in forms from *klh*, *pnh*, and *rbh* in which the initial consonant is not *y*; and also in *yeṣel* (Job 27:8).

shewa in *hyh*, *hyh*.

hireq elsewhere, including monosyllabic forms from *yph*, *pth*, *ṣbh*, and also *wṣz* 2 Kings 9:33, and in forms from *pnh* and *rbh* with initial *y*. The initial consonant clearly influences the development of the vowel in forms from *pnh* and *rbh*, and also *rḥ*. Evidently *y* allows the change to *a* but generally inhibits "tonic lengthening" while *t* allows tonic lengthening. Where the middle (first root) consonant is a guttural, the maximum change is induced (*pataḥ*, *qameṣ* sometimes in pause). Where the final (second root) consonant is a guttural, "tonic lengthening" is usually induced.

5. Under Philippi's law, the *i* vowel of nouns of form **CiCC* should have changed to *a*, as stated by Bauer and Leander. However such "laws" of change do not reflect sudden events but gradual processes, which are facilitated under some conditions and

²⁴ *Bz*, *bt*, *pt*, *sp* (cf. *mwrg*). *Qameṣ* appears in pause in *bz* Num. 31:32, *sp* Exod. 12:22 (and 2 Kings 25:18 in some sources but not in L.). Elsewhere *pataḥ* is shown in pause, as *bz* Isa. 33:23, *bt* Gen. 30:21, *sp* Judg. 19:27.

²⁵ *hk*, *hs*, *ht*, [◌]*ḥ*, [◌]*t*, *kn* "base" and "gnat," *ns*, *ns*, *sn*, *ṣṣ*, *tl*, *tl*.

²⁶ No doubt largely due to the fact that the vowel is never in an open syllable. Cf. the difference between the reflexes of original short *a* in nouns of form **CaC*: (as *kp*) and **CaCC* (as *ksp*). *Pataḥ* is a

fairly common reflex of *i* in what is in MT a closed final syllable in a construct noun.

²⁷ The original prefix vowel of these forms may well have been *a*, but the change to *i* is of such long standing that the patterns shown by the reflexes are similar to those shown by reflexes of *i* elsewhere, even in pause, where the reflex may be *pataḥ*.

²⁸ *hlh* 2 Kings 1:2, [◌]*lh* 2 Chron. 36:23 (the cases in Gen. 24:16, 1 Sam. 2:6, are presumably *hif'il*), *ṣh* Job 23:13. *Pataḥ* appears, for example, in *hrh* Gen. 16:4.

but less favorable conditions had prevented the change to *qameš* up to the time when the development of distinctive pausal forms ceased (§§3.3b, 3.4b, 3.5b).³²

6.4 Where conditions were particularly unfavorable for the change from *i* to *a*, this change did not occur before the syllable became open. This seems to occur chiefly with voiceless continuants, suggesting that, where it occurred, ^o was not a stop, and ^c was not voiced. When the opening occurred, the unchanged vowel gave rise to *šere*, as is usual with original short *i* in open stressed syllables. This has generally occurred even in pause in the forms listed in §§ 3.1 and 3.2. Elsewhere, the change to *a* had always taken place in pause before the opening of the syllable, so that *šere* occurs only in contextual situations. In fact, in many cases the change to *a* had occurred in these situations also, where the word was not closely related to what followed, so that *šere* is generally restricted to construct and other close-joined situations (§3.5a). However, the extent to which *šere* occurs differs in different words, and the information is not adequate to show the factors which condition such small details of usage. Examples are: in *šbr* "breaking," *šere* is used in some construct forms (Amos 6:6, Isa. 30:14), *segol* in others (Lev. 21:19, 19, and eight others). *Segol* is used in other absolute forms, *qameš* in pause (Prov. 17:19, Lam. 3:47). In *sml*, *šere* is used in construct situations (Ezek. 8:3, 5), *segol* in absolute (2 Chron. 33:7, 15), *qameš* in pause (Deut. 4:16). In *škl*, *šere* is used in construct forms, before an adjective, or in the first of a pair of nouns (Ps. 111:10, Prov. 3:4, 13:15, 16:22, 19:11, 23:9, 1 Chron. 22:12, 2 Chron. 2:11, 30:22; see n. 19 above). *Segol* is used at the end of a construct phrase (1 Sam. 25:3, Ezra 8:18) or of a clause (Neh. 8:8, 1 Chron. 26:14). *Qameš* is used in pause (Job 17:4). In *yš^c*, *šere* is used within clauses, including construct situations (Hab. 3:13, 13, Ps. 12:6, 20:7, 50:23), *segol* at the ends of clauses, including pausal situations (Isa. 45:8, 61:10, Ps. 132:16, Job 5:4, 11). It is noteworthy that in these examples, the more limited use of *šere* correlates with use of *qameš* in pause and vice versa, showing that the decisive factor in the development of the vowelings of these forms was the extent to which the change from *i* to *a* had taken place, which varied in each form.

7. The vowelings of these nouns in manuscripts with Babylonian pointing, which differs somewhat from that described above, can be briefly described as follows.³³

7.1 In an open stressed syllable in context, *šere* occurs after initial *alef* in all six cases. With other gutturals, *šere* occurs after *he* in two of three cases, after *het* in five of twelve cases, after *ayin* in six of sixteen cases (of which two others show both *šere* and *pataḥ*). After other consonants, *šere* occurs only in *spr* and *tš^c* (but not in *sbl*, *sml*, *spl*, *str*, *thn*, *tq^c*). Elsewhere, *pataḥ* is used.³⁴ That is, *pataḥ* occurs in the Babylonian

³² That it did cease is shown by the fact that pausal forms are not always used in pausal position, i.e., pausal or contextual forms were fixed according to one convention of reading and did not change when the convention changed. See my article, "Pausal Forms in Biblical Hebrew: Their Function, Origin, and Significance," *JSS* 25 (1980): 170.

³³ The basic information is given in I. Yeivin, *The Hebrew Language Tradition as Reflected in the Babylonian Vocalization* (Jerusalem, 1985), 71 (pp. 808 ff.) I am most grateful to Professor Yeivin for providing me with a copy of this material before its publication. The Tiberian names are used there,

and here, for the Babylonian vowel signs. *Qameš*, *šere*, and *hireq* have a similar range of use in the two traditions. However, the Babylonian *pataḥ* corresponds both to Tiberian *pataḥ* and to most uses of *segol*. Only nouns already studied in the Tiberian tradition are considered here.

³⁴ This describes the standard forms. A few cases where *šere* is used in late and possibly Tiberianizing sources (*nsk*, *sl^c*, *rb^c*) are ignored, as are the few cases where these words show *o* or *u* in any of the three situations, a specifically Babylonian phenomenon.

7.5 The fact that the Babylonian forms do not differ greatly from the Tiberian presumably reflects the development common to the two traditions before they grew apart. The differences which do occur can reasonably be attributed to differing developments in the two traditions after they had grown apart. However, not only do the Babylonian forms provide no reason to question the development of the Tiberian forms suggested in §6, but they appear to support the argument that the different vowels used in situations (a) and (b) reflect the different rates at which the *i* to *a* change developed in different environments.

8. Conclusions.

8.1 The development of *i* in nouns of form **CiCC* is conditioned by those forces which cause natural sound change. There is no need to invoke "analogy" to explain these changes. The factors which can be shown³⁸ to have conditioned the development are: the consonantal sounds which make up the word, particularly that preceding the vowel; the structure (open or closed) of the syllable containing the vowel; and the relation of the word to the surrounding context.

The demonstration of the extent and significance of the influence of this last factor is the most interesting and important result of this study. It shows that the conditioning forces which produce distinctive construct forms or pausal forms are no different from those which affect forms in other situations. The relevant conditioning factors (close-joining in speech of the word in question to what follows to indicate close syntactic/semantic relationship or its separation from what follows to indicate the absence of such relationship) have their maximum effect in construct and in pausal situations. Construct forms and pausal forms represent the extremes of development of any particular process of vowel change.³⁹ However, the developments which lead to these changes in construct or pausal situations also take place in other situations. Here also they are conditioned by the extent to which the word in question is joined to, or separated from, what follows, although they may not lead in these other situations to the same degree of change as they bring about in a construct form or in a pausal form.

8.2 The operation of this factor, the relation of the word to its context, appears to play the major role in determining the appearance of *sere* in groups 3.4 and 3.5 and of *segol* at least in *smi* in 3.2. The operation of this factor, and so the use of these vowels,

pausal forms derive from a base **milk*, and those with suffixes from a base **malk*. The latter would seem the more likely.

³⁸ Other factors, which have left no identifiable trace in the text, may also have been at work. This includes the possibility that some inconsistency may have been caused by the introduction of forms which developed in different strands of the tradition. See Fernández-Tejero, ed., *Estudios Masoréticos*, p. 38.

³⁹ This conflicts with the view that where nouns ending -CVC in MT show different final vowels in absolute and construct states, the difference is due to the fact that the syllable was closed (due to the loss of the case ending) in the construct form earlier than in the absolute. Thus, recently, E. E. Knudsen, "The Mari Akkadian Shift *ia* > *e* and the Treatment of ה"ל Formations in Biblical Hebrew," *JNES* 41 (1982): 37-38, quoting, among others, J. Blau, who

in "Non-Phonetic Conditioning of Sound Change and Biblical Hebrew," *HAR* 3 (1970): 13 applies the same argument to the final vowel of verb forms. This can, indeed, explain the reflexes of *a* in verb forms, and in most nouns (but not all; see again my article in *HAR* 5, p. 98). The argument would, however, require *patah* as the reflex of *i* (through Philippi's law) in the final syllable of construct nouns (where *sere* is at least equally frequent) and in verb forms (where *patah* is common only in originally closed syllables, and is far from consistent there). Consequently, it seems preferable to maintain my argument (*ibid.*) that whatever may have been the earlier history of these forms, the most important factor conditioning the MT reflexes of these vowels is that of "relation to context" as expressed through close-joining or separation in speech and possibly other suprasegmental features.